

NABL Lab Report Document Intelligence

Pipeline Report: Dataset Labeling → Benchmark

Status as of 2026-09-02. Every number in this document was read from a real output file produced by running the actual scripts against the actual 53-file dataset at G:\Shared drives\Product & Engineering\Projects\NABL Document Intelligence\dataset — nothing here is a projection.

Owner: NABL Document Intelligence engineering

Scope: app/dataset_normalization → app/kv_extraction → app/schema_discovery → app/master_schema → app/labeling → app/quality_control → app/final_dataset → app/final_dataset_qc → app/benchmark (backend/)

1. Overview of this document

This report covers everything downstream of the OCR/PDF integration work: a batch pipeline that turns a folder of raw, heterogeneous NABL lab reports into a labeled, structured dataset, and a benchmark that re-runs the production extraction path fresh and scores it against that dataset. It documents eight stages, in order, each with the real counts its last run produced, and states plainly which parts have and have not actually been executed.

The single most important fact to read this report with: the pipeline's own 8th stage generates the labels, and its 6th stage (quality control) is a rule-based filter over those same LLM-generated labels — there is no human-annotated ground truth anywhere in this dataset yet. The benchmark numbers in Section 10a measure whether the production pipeline can reproduce the labeling pipeline's own output, not accuracy against an independently verified answer key. See Section 9 (FAQs).

2. Goal

Build and prove out an automated path that takes any lab report — medical, food, milk, or chemical testing, born-digital PDF, scanned PDF, or photographed image — and produces structured, per-domain key-value test results, without hand-writing a schema for each report template in advance. Then measure, with a real benchmark run, how well the production extraction code actually achieves that.

3. Requirements

3a. Functional

- Accept PDF, JPG, JPEG, PNG, and TIFF input regardless of whether the source is born-digital or scanned.
- Assign every document a stable, content-hash-derived ID (LR_NNNNNN) so reruns and renamed files never get a second ID, and byte-identical duplicates are detected automatically.
- Decide born-digital vs. scanned *per page*, not per document, and fall back to OCR only on pages that need it.
- Discover each domain's real field vocabulary from the documents themselves, before any fixed schema exists.
- Merge discovered fields into one master schema per domain, plus a shared "common" bucket for fields that recur across domains.
- Classify each document's domain and extract its structured fields/tests in one pass, guided (not constrained) by the master schema.
- Gate every label through rule-based quality checks before it is allowed to count as ground truth.
- Assemble one canonical final dataset (JSONL + CSV) from only the approved labels.
- Independently re-validate the assembled dataset's integrity, as a separate read-only pass.
- Benchmark the real production pipeline by re-running it fresh against the raw files and scoring the result against the assembled dataset.

3b. Non-functional

- Raw source files are never modified; every stage writes to its own separate output directory.
- Every stage is resumable — already-processed documents are skipped on rerun, with a --force flag to override.
- One bad or corrupt document must never abort a batch run.

- No new OCR or LLM implementation: reuse the existing deepreef-ocr integration (`app/documents/local_ocr.py`) and the existing multi-provider LLM chain (`app/llm/factory.py`) exactly as the live app already does.
- No metric or count is reported unless it came from an actual completed run — an unrun stage is reported as unrun, not estimated.

4. Background

This pipeline follows on from getting English/Latin OCR genuinely wired into the NABL backend (reusing the sibling deepreef-ocr repository directly, not a vendored copy) and from evaluating two public HuggingFace lab-report datasets as candidate inputs. Neither public dataset ended up feeding this pipeline: the 53-file internal dataset at the path above (real, redacted NABL client reports spanning medical, food, milk, and chemical testing) was used instead, since it matches the reports the production system will actually see.

5. Who's your customer

- **Primary:** the NABL Document Intelligence engineering team, who need a trustworthy, repeatable way to answer "how accurate is the extraction pipeline right now" before changing it.
- **Secondary:** any product surface that consumes the structured output — e.g. the NABL backend's document upload/extraction flow — benefits from the master schema and QC rules this pipeline produces as reusable artifacts, not just a one-off report.

6. Tenets

- Real data over synthetic data — every stage below has been run against the actual 53-file dataset, not fixtures.
- Reuse, don't reimplement, existing OCR/LLM infrastructure.
- Rule-based before LLM wherever a heuristic can do the job (schema clustering, quality control, and the standalone table-row extractor are all LLM-free).
- Every stage's output lands on disk as inspectable JSON/JSONL/CSV, not hidden inside a database.
- Report what actually ran. An absent result is reported as absent, never backfilled with a plausible-looking number.

7. Design

7a. Challenge

No two lab report templates share a layout. The column that holds the acceptance limit comes last in some templates and second-to-last in others; some reports are clean born-digital PDFs and others are photographed or scanned; the set of domains (medical, food, milk, chemical, ...) and the fields each one cares about aren't known ahead of time; and there is no pre-existing, human-verified answer key to test against — one has to be bootstrapped from the documents themselves before any accuracy number is even meaningful.

7b. Design options considered (schema & extraction strategy)

Option	Approach
A — Rule-based only	Regex/heuristic table-row parser reading numbered rows straight out of the flattened page text (built first, shipped as the standalone <code>kv_extraction</code> stage).
B — Fixed schema + LLM	Hand-author one field list per domain up front, then have the LLM fill it in for every document.
C — Discovered schema + LLM (chosen)	Sample real documents per domain with an LLM to discover the fields that actually appear, cluster them into one master schema per domain, then run a second LLM pass per document guided by that discovered schema.

7c. Pros

- **A (rule-based):** zero LLM cost or latency, fully deterministic, trivial to unit-test.
- **B (fixed schema):** simplest to build; predictable field names from day one.
- **C (discovered schema, chosen):** field vocabulary reflects what real reports actually contain instead of what an engineer guessed; a document from a domain no one anticipated still gets *a* domain and best-effort fields; the master schema and its 69 fields shared across ≥ 2 domains become a reusable artifact for later work.

7d. Cons

- **A (rule-based):** template-specific — needed four separate bug fixes to handle two real reports' column-order and row-numbering differences (Section 10c), and still only found rows in 32 of 50 documents; it does not classify domain or handle free-form (non-tabular) fields at all.
- **B (fixed schema):** brittle against any report the schema author didn't anticipate; misses real fields that exist in the documents but weren't guessed in advance.
- **C (discovered schema, chosen):** two LLM calls per document lifecycle (discovery sample + label) instead of zero or one; schema discovery only sampled 23 of 50 documents, so rarer domains (only 1–2 sampled documents each) have a thinner basis than medical/food; the "ground truth" it produces is LLM output passing rule checks, not human-verified (Section 9).

Chosen architecture

#	Stage	Package	LLM?	Last real run
0	Ingest	—	No	53 raw files found
1	Normalize	<code>dataset_normalization</code>	No (OCR only, existing)	50/50 processed, 0 failed

#	Stage	Package	LLM?	Last real run
1b	Table-row extraction (standalone)	kv_extraction	No	32/50 docs, 259 rows
2	Schema discovery	schema_discovery	Yes	23/50 docs sampled
3	Master schema	master_schema	No	559 cluster entries
4	Labeling	labeling	Yes	50/50 labeled
5	Quality control	quality_control	No	50/50 approved
6	Final dataset	final_dataset	No	991 fields, 601 tests
7	Final dataset QC	final_dataset_qc	No	50/50 valid (run 2026-09-02)
8	Benchmark	benchmark	Yes (predict only)	50/50 predicted & scored

Stage 1b (kv_extraction) is not consumed by any later stage — it is an earlier, independent rule-based experiment that reads the same normalized documents but was superseded by the LLM-based schema-discovery/labeling path for anything schema-aware. It is included here because it is real, working, and still in the repository.

8. Conclusion

The full chain is code-complete and has been run end-to-end for real, twice over the same 53-file dataset (once through Section 6, and a second pass on 2026-09-02 that closed out the two stages left unrunk: final-dataset QC and the benchmark's scoring phase). 258 automated tests pass across all eight packages, 0 failing.

- 53 raw files → 50 unique documents (3 byte-identical duplicates deduplicated by content hash) → all 50 normalized with 0 failures.
- 50/50 documents labeled by the LLM pass; 50/50 passed rule-based QC and are marked approved.
- Final dataset: 991 total fields, 601 test rows, 0 documents excluded by validation.
- Final-dataset QC (independent re-check, run 2026-09-02): 50/50 structurally valid, 0 duplicates, 0 invalid annotations — but 108 unexpected keys and 413 missing values surfaced (Section 10a), i.e. schema drift that the earlier stages' own checks don't catch.
- Benchmark (fresh OCR + fresh LLM call per document, scored against the final dataset, run 2026-09-02): overall key-level F1 **0.832** (precision 0.821 / recall 0.843), field accuracy **84.5%**, exact-match rate 22%, hallucination rate 17.9%.

The weakest results are concentrated, not spread evenly: the **medical** domain (14 documents) and **image**-format documents (10 documents, all OCR'd) both show a hallucination rate above 44% and key precision below 0.56, well behind every other domain/format. Medical reports carry the widest field vocabulary discovered (218 keys) from the thinnest discovery sample (7 documents), which is a plausible cause worth investigating before trusting this pipeline on medical reports specifically.

9. FAQs

Q: Is the benchmark score an accuracy number I can quote externally?

No, not yet, and this is the single most important caveat in this report. The "ground truth" it scores against (`final_labeled_dataset.jsonl`) was itself produced by an LLM (Stage 4), filtered only by rule-based checks (Stage 5) with no human review step anywhere in the code. The benchmark therefore measures whether the same-family LLM call, run fresh in the production path, reproduces the labeling pass's own earlier output — a reproducibility/consistency signal, not accuracy against an independently verified answer.

Q: Why do two different key-value extraction stages exist?

Stage 1b (`kv_extraction`) was built first, is rule-based, and works well on documents with a clear numbered results table (32/50 documents), but has no notion of domain or free-form fields. Stages 2–4 (schema discovery → master schema → labeling) replaced it as the schema-aware, all-50-document path used everywhere downstream. Stage 1b was left in place rather than deleted because it is real, tested, working code.

Q: What happens to a document that fails to process?

Every stage is designed so one bad document is recorded with an error and status, and the batch continues. In this dataset's actual runs, zero documents failed at any stage (normalization: 0/50 failed; labeling: 50/50 `extraction_status="ok"`; final dataset: 0 excluded by validation) — that guarantee has not yet been exercised by a real failure in this dataset.

Q: Why did final-dataset QC and the benchmark's score phase show as "not run" until 2026-09-02?

They were built and unit-tested but their CLI entrypoints (`validate_final_dataset.py`, and the score half of `run_benchmark.py`) had not been executed against the real dataset before this report was assembled. Both are read-only/re-scoring passes with no LLM calls in the QC case, so running them to produce this report added no new risk to any existing output.

Q: Does the pipeline handle non-English or handwritten reports?

Out of scope for this report and this dataset. The OCR path used throughout (deepreef-ocr's English model) is the same one integrated earlier in this project; no handwriting-specific handling exists.

Q: Can this scale past 50 documents?

Architecturally yes — every stage is a batch loop over a directory with resumable, per-document caching. Not yet demonstrated: schema discovery only sampled 23 documents here, and the **water** and **soil** domains have zero real documents in this dataset, so their master-schema entries and QC rules are entirely unexercised.

10. Appendix

10a. Metrics — benchmark results (2026-09-02 run, 50/50 documents)

Scope	n	Key P	Key R	Key F1	Field acc.	Exact match	Halluc. rate
Overall	50	0.821	0.843	0.832	0.845	0.220	0.179
chemical	7	0.904	0.867	0.885	0.881	0.286	0.096
food	18	0.926	0.889	0.907	0.873	0.389	0.074
medical	14	0.560	0.753	0.642	0.829	0.143	0.440
milk	9	0.811	0.819	0.815	0.793	0.000	0.190
other	2	0.375	0.429	0.400	0.619	0.000	0.625
born_digital_pdf	35	0.876	0.874	0.875	0.867	0.286	0.124
scanned_pdf	5	0.679	0.667	0.673	0.663	0.000	0.321
image	10	0.532	0.733	0.617	0.840	0.100	0.468

water and soil domains: 0 documents in this dataset, all metrics null — omitted above. Source: `benchmark/benchmark_results.json`.

Final-dataset QC (independent integrity re-check, 2026-09-02)

Check	Result
Documents checked	50
Structurally valid	50 / 50
Duplicate documents	0
Invalid annotations	0
Unexpected keys (not in master schema)	108
Missing values (field present, value null/empty)	413

10b. Performance

Stage	Measured
Normalization	10.44 s/document average (OCR-bound pages dominate); 138 total pages across 50 documents.
Schema discovery / Labeling / Benchmark predict	Per-document wall time not logged by these scripts; each involves one network LLM call plus, for benchmark predict, a fresh OCR pass.
Quality control / Final dataset / Final dataset QC / kv_extraction	No LLM calls; rule-based, seconds for the full 50-document batch.

10c. Edge cases handled (with real examples from this dataset)

- **Byte-identical duplicate files** — 3 of the 53 raw files hash to an ID already assigned; they are skipped rather than reprocessed or given a second ID.
- **Mixed born-digital/scanned PDFs** — 2 documents have some pages with a real text layer and others without; each page is OCR'd or not independently.

- **Password-protected PDFs** — detected via PyMuPDF's `needs_pass` before any text extraction is attempted, raised as a clear error rather than silently returning empty text.
- **Row-number format variance** (`kv_extraction`) — one real report numbers rows "1." with a trailing period instead of "1"; the parser matches both.
- **Column-order variance** (`kv_extraction`) — one real report's table ends in a limit column, another's ends in the result itself with no limit column; the parser reads each document's own header row to decide which convention applies, per document.
- **End-of-table runaway** (`kv_extraction`) — without a row-count cap, the last numbered row would swallow trailing signature/remarks text with no natural terminator; a max-cells-per-row cap stops it.

10d. Glossary

Term	Meaning
LR_NNNNNN	A document's stable ID, derived from a sha256 hash of its raw bytes, assigned once and reused on every rerun regardless of filename.
Domain	One of medical / food / milk / chemical / water / soil / other — the category of testing a report represents, assigned by the labeling LLM call.
Master schema	The merged set of field names per domain, built by clustering what schema discovery found across sampled real documents into one canonical list.
Common key	A field name whose cluster appears in 2 or more domains' discovered vocabularies (69 such keys), kept as one shared entry rather than duplicated per domain.
Approved (QC)	A label that currently passes every rule-based check in <code>app/quality_control/checks.py</code> . Not human-reviewed — see Section 9.
Key precision / recall / F1	Of the field+test keys the production pipeline extracted for a document, how many match the final dataset's keys (precision), how many of the final dataset's keys it found (recall), and their harmonic mean (F1).
Field accuracy	Of the keys both sides agree exist, the fraction whose values also match.
Exact match rate	Fraction of documents where every field and test value matched the final dataset exactly, with zero discrepancies.
Hallucination rate	Fraction of the keys the production pipeline extracted that do not appear in the final dataset at all.
Born-digital / scanned / mixed PDF	A PDF whose pages carry a real, selectable text layer / have no text layer and need OCR / contain a mix of both.